

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2013–14

Paper: CSB-603A — System Analysis and Design

Time: Three Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. Answer all parts:

[3 + 3 + 4 + 4 = 14]

- (a) What is the difference between a program and software? Explain.
- (b) Write the purpose of CASE tools.
- (c) What is system testing? Explain.
- (d) What are the myths associated with software?

Question 2.

[7 + 7 = 14]

- (a) Compare the Waterfall model with the Spiral model of software development.
- (b) What are fact-finding techniques and where are they used?

Question 3.

[10 + 4 = 14]

- (a) An airline reservation is an association between a passenger, a flight, and a seat. Select a few pertinent attributes for each of these entity types and represent a reservation in an E-R diagram.
- (b) What is the difference between a decision tree and a decision table?

Question 4.

[6 + 8 = 14]

- (a) Differentiate between the terms: Data Flow Diagram (DFD), Flowchart, Data Dictionary, and E-R diagram.
- (b) Define module coupling and explain the different types of coupling.

Question 5.

[4 + 10 = 14]

- (a) What is the difference between testing and debugging? Explain.
- (b) Write in detail about all types of software testing methods.

Question 6.

[2 + 12 = 14]

- (a) What is the difference between verification and validation?

(b) Write short notes on the following:

- (i) Modular top-down analysis
- (ii) Maintenance process
- (iii) User interface design

Question 7.

[5 + 4 + 5 = 14]

- (a) What is Object-Oriented Analysis and modelling?
- (b) What are the characteristics of a Software Requirements Specification (SRS)?
- (c) Write about the significance of the coding phase in SDLC.